5 **(a)** Describe an experiment to determine the speed of sound in air.

State the apparatus required, the measurements taken and how to calculate the speed of sound in air from the measurements.

You may draw a **labelled** diagram as part of your answer.

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[4]

(b) Bluetooth is used to connect a mobile (cell) phone to headphones.

(i) State the name of the region of the electromagnetic spectrum used by Bluetooth.

..... [1]

(ii) State the speed of electromagnetic waves in air.

speed = ... [1]

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(iii) The frequency of a Bluetooth network is 2.48 GHz.

Calculate the wavelength of the waves used in the Bluetooth network.

wavelength = ... [3]

[Total: 9]

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